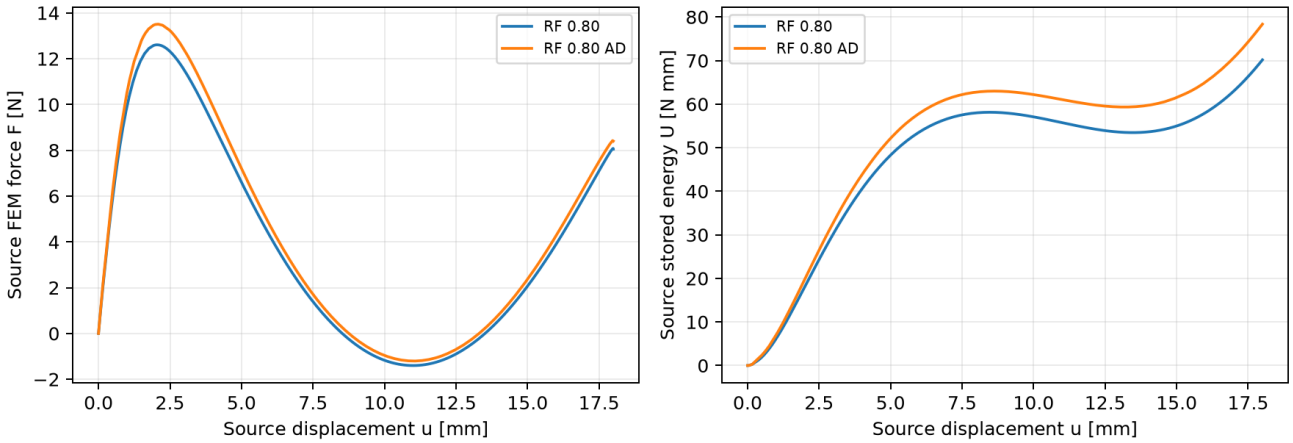
Kresling origami: native CAD and nonlinear FEM

12 October 2026. Three CAD/FEM source projects added: compliant origami, power-plant valve thermal conduction and two connecting-rod load cases. An additional mounting-flange drawing/scalar source is held for drawing reconstruction. New broad families are origami mechanisms and industrial valve thermal analysis; rod/flange extend existing mechanical families.



Original stored CAD preview: PNG extracted unchanged from the Inventor thumbnail property. Original native IPT and binary Parasolid X_B are retained. This small source preview is not an independently imported STEP view or a dimensioned drawing. Inventor template filename says n8; selected Parasolid/deck names identify n6. Exact template parameters need review.

Read-only plot of published FEM workbook values; no new solve



Read-only plot of existing FEM cell values: workbook RF 0.80 DM60, A7:C217 and E7:G107. Displacement, force and stored energy are copied from source FEA columns; EXP data is excluded. RF 0.80 has 211 samples, peak force 12.604324 N and minimum -1.401440 N. The AD-labeled curve has 101 samples and peak 13.505147 N. No new FEM values or energy integration.

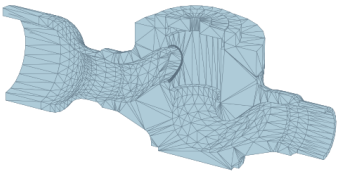
36 published FEM curves are retained across two workbooks with exact cell columns and row IDs. They cover reduction/material/time variants, not 36 downloadable CAD pairs. One RF0.80 CAD-linked design candidate is registered; exact curve/deck mapping remains pending.

Example Abaqus deck uses geometric nonlinearity, viscoelastic creases, elastic-plastic panels, a quasistatic VISCO step of 18 s and imposed -18 mm Y displacement. CAD is scale x1; source example mesh is x3. No CAD scaling or source program was executed. No ODB/full nodal fields located.

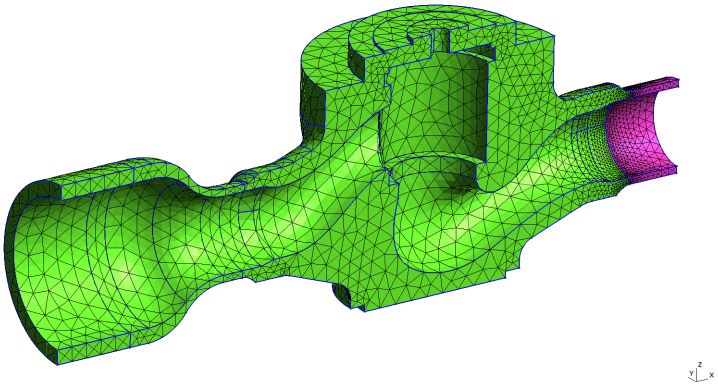
Credit: Samantha Mora, Nicola Pugno and Diego Misseroni, University of Trento; DOI 10.5281/zenodo.15118000, CC BY 4.0. Native geometry import, exact material/amplitude mapping and convergence need validation.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Industrial valve: original BREP and thermal setup



Read-only source CAD inspection: original valve.brep independently imports as a valid BREP with four solids and 112 faces. It represents the symmetric valve/nozzle/pipe region. Display triangles are CAD tessellation, not a newly generated FEM mesh. No dimensioned manufacturing drawing is claimed.



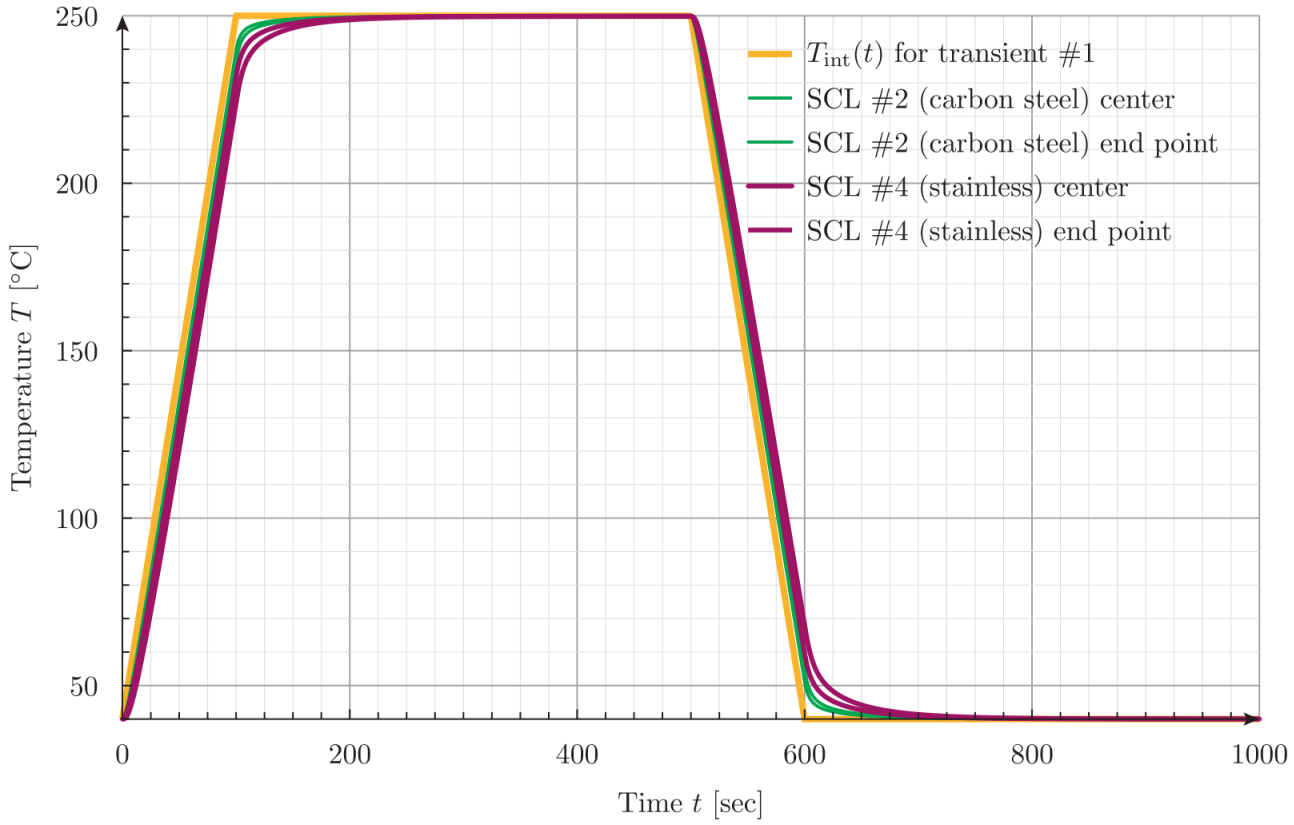
Original publisher material view: stainless valve joined to carbon-steel piping. Source .geo defines volume and boundary labels; .fee defines nonlinear conductivity/diffusivity and transient thermal conditions. Scripts are retained as text and were never executed.

Model evidence	Published setup
Internal boundary	40 C at 0 s, 250 C at 100/500 s, 40 C at 600/1000 s; linear interpolation
External / symmetry	Convection h=1e-6 W/(mm2 K) to 50 C; symmetry adiabatic
Materials	Carbon steel and stainless, temperature-dependent source property tables
Initial state	Source solver starts with the steady-state solution at t=0

This is one complex valve thermal candidate. Six sampled locations and 192 time steps remain one geometry task. Source BREP-to-volume/surface/SCL mapping, original mesh correspondence and mesh/time convergence require review.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Valve: existing numerical FEM temperature histories



Original published FEM plot: rasterized from source SVG. Source CSV has 192 rows, 0-1000 s. Column 2 is imposed internal temperature and is excluded as a solved reference target. Columns3-8 are the six published FEM temperatures on carbon/stainless stress-classification lines (SCL).

Source FEM location	CSV column	Peak C	Final C at 1000 s
SCL2 carbon inner	3	250.0	40.0
SCL2 carbon center	4	249.926	40.011
SCL2 carbon outer	5	249.863	40.018
SCL4 stainless inner	6	250.0	40.0
SCL4 stainless center	7	249.913	40.023
SCL4 stainless outer	8	249.844	40.038

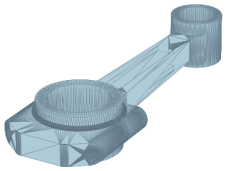
Original-temperature-history.npz preserves the original time vector, imposed boundary table/values and all six sampled FEM temperature columns. Source coordinates: SCL2 x=+3 mm and SCL4 x=-3 mm, y=0, from z=48.5 to 57 mm with center at 52.75 mm. Coordinates and physical material sides require explicit topology matching.

Values are source-rounded to 0.001 C. Original full nodal mesh/valve-1.msh result file is not included in this source repository. The CSV supplies real FEM trajectories; it is not a complete nodal-field benchmark. No fresh mesh or solve was created.

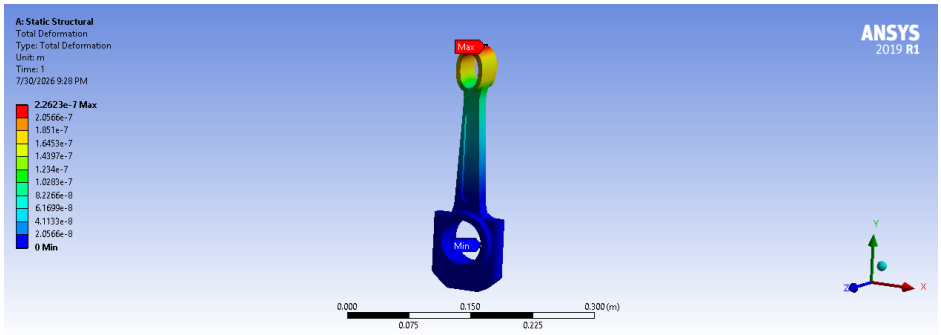
Compare model trajectories at the same source points and times after resolving mesh and units. Reproducing the imposed boundary is not a measure of solving the interior thermal problem. Zero certified admissions or measured frontier-model scores.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Connecting rod: original CAD and two saved results



Read-only original neutral CAD: connecting_rod.STEP is valid, with two solids and 76 faces, approximately 94 x 283.7 x 38 mm. Archive contains native SolidWorks rod CAD, two SpaceClaim documents, complete source decks/logs and two original RST files. Full engine native assemblies/parts and Sheet-1.SLDDRW are also preserved; their dependencies and drawing dimensions are unverified.



Original ANSYS FEM deformation: peak 2.2623e-7 m. Reader extraction matches 2.2623073e-7 m =0.000226231 mm. Companion original stress plots show 5.2316e5 Pa, consistent with recovered peak523,163.39 Pa =0.523163 MPa.

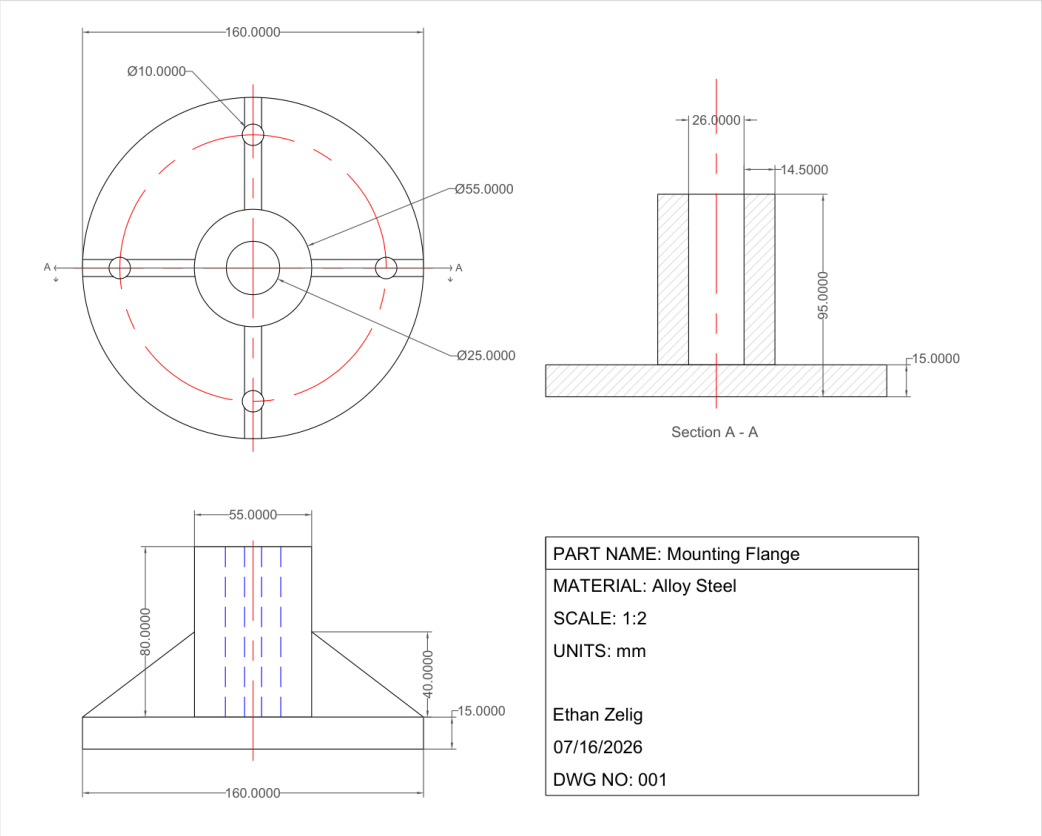
Saved source case	Nodes / elements	Max U mm	Peak averaged VM MPa
Tensile	27,646 /17,170	0.000226231	0.523163
Opposite signed loading	27,646 /17,170	0.000226231	0.523163

Full source displacement and stress arrays are negatives of each other after matching IDs, with identical missing-stress masks. Two opposite linear-static cases remain one rod geometry; they do not establish nonlinear compression/buckling or a solved full-engine assembly.

Source decks use /units,MKS and steel E=200 GPa, nu=0.3. Applied surface-traction card component 2 is +/-12,391.413196 Pa. Saved stresses are finite at 4,238 nodes and missing at 23,408; NaNs/masks are preserved. Exact neutral/native transform matching and convergence remain pending.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Mounting flange: original dimensioned CAD drawing



Original publisher engineering drawing: one-page AutoCAD-exported PDF, with top/front views, section A-A and a title block specifying millimeters, alloy steel and scale 1:2. This is downloaded drawing evidence. No drawing or flange CAD was created locally.

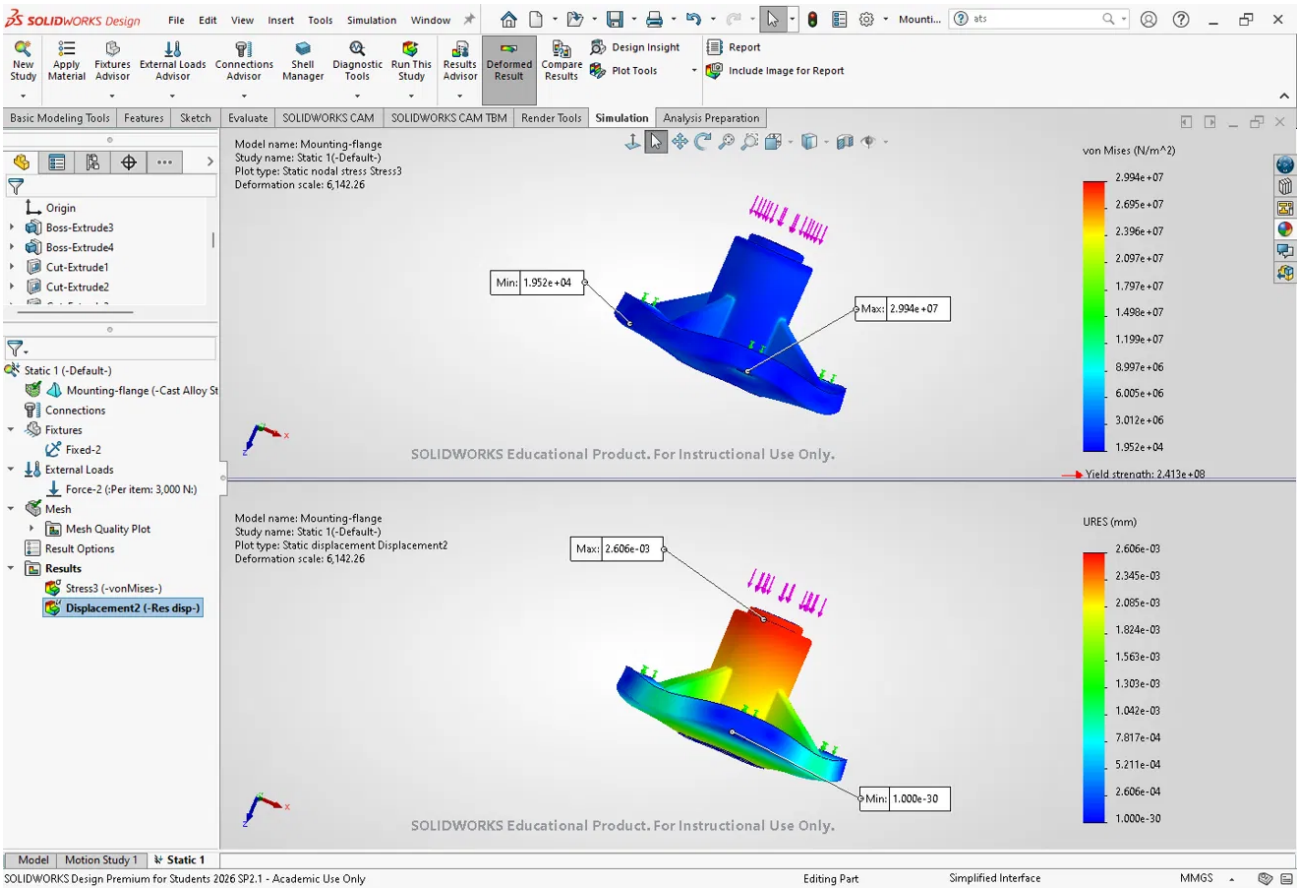
Visible/source dimension	Review finding
Flange OD 160; thickness 15; hub OD 55	Dimensioned on original drawing
Bore	Plan labels 25 mm; section labels 26 mm; README says 26 mm
Hub height	Front 80 mm; section 95 mm needs interpretation
Bolt pattern / ribs / fillets	README supplies Ø130 mm PCD and 8 mm ribs; drawing lacks complete variable-fillet specification

One drawing-based FEM candidate is held. No SLDPRT/STEP model or full saved SolidWorks result binary was uploaded. Drawing alone is insufficient to certify all geometric details. It does not provide 55 original CAD models for the 55 scalar simulation rows.

The original native engine SLDDRW from the rod project is also retained, but has not been independently rendered or dimension-verified. One newly verified dimensioned engineering drawing file in this extension; zero complete manufacturing packages verified.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Mounting flange: published FEM scalars and limits



Original source SolidWorks Simulation figure: source applied load 3000 N, peak stress 2.994e7 Pa =29.94 MPa and URES 0.002606 mm. Source deformation display scale 6,142.26 exaggerates shape. The screenshot is preserved unchanged.

Fillet mm, load 3000 N	CSV stress MPa	CSV U mm
1	3.022E+01	2.608E-03
2	2.940E+01	2.577E-03
3	3.080E+01	2.546E-03
4	3.066E+01	2.514E-03
5	3.018E+01	2.479E-03

The figure stress and displacement do not simultaneously match a single table row. CAD/fillet/setup revision and screenshot scope are unresolved. Preserve source figures and CSV values without silently replacing, fitting or rescaling them.

CSV records 55 claimed linear-static runs: 11 loads x 5 fillet radii, with stress, displacement and safety factor. It is one drawing/scalar campaign, not 55 CAD/reference pairs. Original-source-scalar-table.npz exactly retains 55 IDs and numerical rows.

Source fixtures fix the four bolt-hole cylindrical faces and apply distributed normal force to hub top. Exact solver material, mesh, fixture/load mapping and convergence need confirmation. Published surrogate-network accuracy is not a frontier-model benchmark score.

Source: original publisher project / dataset. Original URLs, revisions and SHA-256 provenance retained.

Collection inventory and benchmark preparation

New source evidence	Count / role
Kresling origami	1 native CAD-linked curve candidate; 36 auxiliary FEM curves; original IPT/X_B and Abaqus deck
Power-plant valve	1 valid native BREP with 6 FEM temperature trajectories; 192 source time samples
Connecting rod	1 valid STEP geometry with 2 original static RST cases; engine CAD/drawing auxiliary
Mounting-flange drawing	1 dimensioned engineering drawing and 55 source scalar rows; no native3D CAD uploaded
Rocket turbopump held	Original STEP downloaded, but structural FEA is future work; CFD excluded
Certified cases / model scores	0 / none

Provenance and checks: all 40 selected original downloads passed publisher MD5 or Git blob SHA-1, byte-size and SHA-256 checks. All 36 FEM curve copies, six valve histories and 55 scalar table rows were compared with the original cells/files. Rod opposite signed field consistency is verified. Original source files, labels, notices and unresolved limits remain visible.

Supplied CAD to FEM: valve and rod have independently imported valid CAD. Origami native CAD needs compatible import and the x1/x3 source correspondence resolved. Freeze exact setup, geometry revision, units, output point/mode conventions and mesh/error tolerances before model trials.

Drawing to CAD to FEM: flange adds useful original orthographic/section evidence, with bore and detail conflicts explicitly held. Verify missing dimensions and original 3D model/result mapping before using it for reconstruction scoring. Native rod drawing remains unrendered.

Held turbopump source: Report.pptx slide 10 lists structural/centrifugal FEA as Next Steps. Slide 9 metrics are CFX pressure, torque and efficiency, rather than saved structural FEM. It is excluded from paired counts despite its complex CAD. No result was invented for it.

Licenses: origami CC BY 4.0, credited to Mora/Pugno/Misseroni; FeenoX GPL3 with original notice; flange MIT; rod and turbopump data-license status unresolved. Local research collection only. Original native files are preserved; no public release or license grant is implied.

Three CAD/FEM source projects. Two new broad family candidates: compliant origami and industrial valve thermal conduction. Rod and flange extend already represented families. Samples, scalar rows, material variants and opposite load directions are not extra families.

No locally generated CAD, source program execution, new FEM solve, COMSOL dependency or frontier-model call. Source references are benchmark candidates; certified admission remains zero.